

Week 3: Introduction to Machine Learning

Task 1 - Machine Learning Fundamentals Report | Name: Rohit Kumar Gupta | AI Internship

1. What is Machine Learning?

Machine Learning (ML) is a part of Artificial Intelligence in which a computer learns from data instead of being given fixed rules by a programmer. In normal programming we write the rules and give the data to get an answer. In machine learning we give the **data and the answers**, and the computer finds the rules by itself. After learning, it can make predictions on new data it has never seen. For example, if we show a model thousands of emails marked 'spam' or 'not spam', it learns the pattern and can then detect spam in new emails.

2. AI vs ML vs Deep Learning

These three terms are related but not the same. Deep Learning is a part of Machine Learning, and Machine Learning is a part of Artificial Intelligence.

Term	Meaning	Example
Artificial Intelligence (AI)	Making machines act intelligently, like thinking, deciding and solving problems.	Chess-playing program, virtual assistants
Machine Learning (ML)	A way to achieve AI where the machine learns patterns from data.	Spam filter, house price prediction
Deep Learning (DL)	A type of ML using multi-layer neural networks, works well on images, speech and text.	Face recognition, ChatGPT-like chatbots

3. What is a Dataset?

A dataset is a collection of data used to train and test a machine learning model. It is usually arranged like a table where each **row** is one record (for example, one student) and each **column** is one property of that record (study hours, attendance, result). Common examples are the Iris dataset, House Prices and Student Performance data. The quality of the dataset directly affects the quality of the model.

4. Features and Labels

- **Features (X)** are the input values the model uses to make a prediction. In a student dataset, Study Hours, Attendance and Previous Marks are features.
- **Label / Target (y)** is the output we want the model to predict. In the same dataset, Result (Pass/Fail) is the label.

5. Training Data and Testing Data

We split the dataset into two parts. **Training data** (usually 70-80%) is used by the model to learn. **Testing data** (20-30%) is kept hidden and is used only after training to check how well the model works on new data. If we tested on the training data itself, the model could simply memorize it (overfitting) and the score would be misleading.

Flow: Training Data → Model Learning → Testing Data → Prediction

6. Supervised Learning

In supervised learning the data has **labels**, so the model learns from examples with correct answers, like a student learning with a teacher. It learns the relation between X and y and then predicts y for new X.

Examples: predicting whether a student passes, detecting spam emails, predicting house prices.

Algorithms: Linear Regression, Logistic Regression, Decision Tree, KNN.

7. Unsupervised Learning

In unsupervised learning the data has **no labels**. The model has to find hidden patterns or groups on its own. Examples: grouping customers by buying behaviour (customer segmentation), grouping similar news articles, finding unusual transactions. Algorithms: K-Means Clustering, Hierarchical Clustering, PCA.

Point	Supervised Learning	Unsupervised Learning
Data	Labeled (X and y)	Unlabeled (only X)
Goal	Predict a known output	Find hidden patterns / groups
Examples	Spam detection, price prediction	Customer segmentation
Common algorithms	Logistic Regression, Decision Tree	K-Means, PCA

8. Classification vs Regression

Both are types of supervised learning. They differ in the kind of output that is predicted.

Point	Classification	Regression
Output	A category / class	A continuous number
Example	Pass or Fail, Spam or Not Spam	House price, temperature, marks
Algorithms	Logistic Regression, Decision Tree	Linear Regression
Metric	Accuracy, Precision, Recall	Mean Squared Error, R2 score

9. Real-World Applications of Machine Learning

- **Email spam filtering:** Gmail classifies emails as spam or not spam (classification).
- **Recommendation systems:** Netflix, YouTube and Amazon suggest movies, videos and products based on past behaviour.
- **Healthcare:** ML helps detect diseases from X-rays and scans, and predicts patient risk.
- **Banking and finance:** Detecting credit card fraud and deciding loan approvals.
- **Self-driving and maps:** Predicting traffic and recognising roads, signs and pedestrians.
- **Voice assistants and translation:** Google Assistant, Siri and Google Translate understand speech and text.
- **Education:** Predicting student performance, as done in my mini project this week.

10. Conclusion

Machine Learning allows computers to learn from data and make predictions without being explicitly programmed for every case. In this week I learned the basic terms (dataset, features, labels), the difference between supervised and unsupervised learning, and between classification and regression. I also understood why data must be split into training and testing sets, and the basic ML workflow: load data, clean it, split it, train a model, predict and evaluate. These basics will help me in learning advanced topics in the coming weeks.